

PCA - PRINCIPAL COMPONENTS ANALYSIS

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GARTNER'S MAGIC QUADRANT (JANUARY 2023)



WHICH KIND OF DATA?

Rectangular data: data tables where rows are considered as individuals and columns as quantitative variables. Some examples:

- Sensory analysis: score for attribute k of product i
- Ecology: concentration of pollutant k in river i
- Economics: indicator value k for year k
- Genetics: expression of gene k for patient i , or tumor i
- Biology: Measure k for animal i
- Marketing: Value of measure k for brand i
- Sociology: time spent on activity k by individuals from social class i

CEREALS DATA SET

```
> cereal.pc[1:25,]  
      cups calor cabs fat fiber pot prot sod sugar  
100%_ 0.33    70  5.0   1 10.0 280   4 130    6  
Al-B   0.33    70  7.0   1  9.0 320   4 260    5  
A-B_   0.50    50  8.0   0 14.0 330   4 140    0  
A-C_   0.75   110 10.5   2  1.5  70   2 180   10  
Ap_J   1.00   110 11.0   0  1.0  30   2 125   14  
Bs_4   0.75   130 18.0   2  2.0 100   3 210    8  
Br_C   0.67    90 15.0   1  4.0 125   2 200    6  
Br_F   0.67    90 13.0   0  5.0 190   3 210    5  
C''C   0.75   120 12.0   2  0.0  35   1 220   12  
Chrs   1.25   110 17.0   2  2.0 105   6 290    1  
C-T_   0.75   120 13.0   3  0.0  45   1 210    9  
Clst   0.50   110 13.0   2  2.0 105   3 140    7  
Cc_P   1.00   110 12.0   1  0.0  55   1 180   13  
Cr_C   1.00   110 22.0   0  0.0  25   2 280    3  
Cr_F   1.00   100 21.0   0  1.0  35   2 290    2  
Cr_P   1.00   110 13.0   0  1.0  20   1  90   12  
Cn_C   1.00   110 12.0   1  0.0  65   1 180   13  
C'_O   0.50   110 10.0   3  4.0 160   3 140    7  
Crsp   1.00   110 21.0   0  1.0  30   2 220    3  
C_W_   0.75   100 11.0   1  2.0 120   2 140   10  
Db_C   0.75   100 18.0   0  1.0  80   2 190    5  
Fr_L   1.00   110 11.0   1  1.0  30   2 125   13  
Fr_F   0.75   110 14.0   0  1.0  25   1 200   11  
F_M-   0.80   100 14.0   0  3.0 100   3   0    7  
F_&_   0.67   120 12.0   2  5.0 200   3 160   10
```

9 variables on different cereal brands

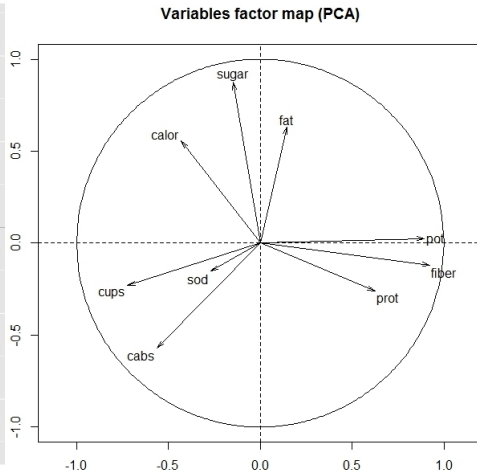
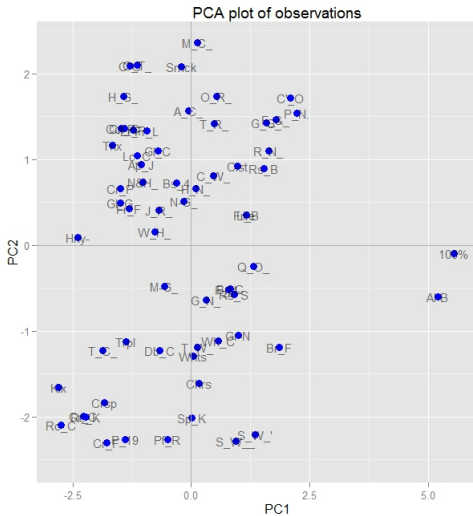
CEREALS DATA SET

By looking at the data, can we say anything about

- Similarities among cereals?
- Relationships between variables?
- Any patterns of variation?
- The global structure of dispersion?

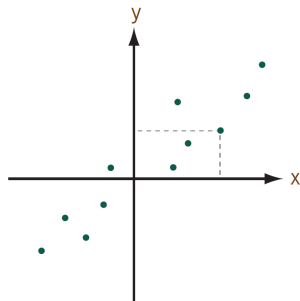
We are searching for a better low-dimensional summary of the data (more informative than traditional EDA). We are interested in studying resemblances between observations (individuals) and relationship among variables.

A PICTURE IS WORTH A THOUSAND NUMBERS...

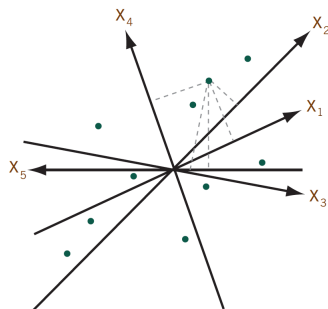


BIPLOTS

They are generalizations of the scatterplot.



Scatterplot



Biplot

► Greenacre's Book: Biplots in practice

With PCA we seek to **reduce the dimensionality** (condense information in variables) of a data set while retaining as much as possible of the variation present in the data.

OVERALL GOALS

- Summarize a data set with the help of a small number of synthetic variables (i.e. the Principal Components)
- Visualize the position (resemblance) of individuals.
- Visualize how variables are correlated.
- Interpret the synthetic variables.

PCA can be used for:

- ① Dimension Reduction
- ② Visualization
- ③ Feature Extraction
- ④ Data Compression
- ⑤ Smoothing of Data
- ⑥ Detection of outliers
- ⑦ Preliminary process for further analysis

PCA FROM DIFFERENT PERSPECTIVES

PCA allows us to

- to **best summarize** the important information contained in a data table
- to find a **graphical representation** of the essential information contained in a data set (through Biplots)
- to find an **optimal approximation** of a data set with a minimal loss of information.

PCA AS BEST SUMMARY OF INFORMATION

Given a set of p variables $X_1, \dots, X_p$, we want to obtain r new variables $Z_1, \dots, Z_r$, called the **Principal Components** (PCs)

Variables

X_1

X_2

⋮

X_j

⋮

X_p



Principal
Components

Z_1

Z_2

⋮

Z_r

PCA AS BEST SUMMARY OF INFORMATION

We want to compute them as linear combinations of the original variables:

$$PC_1 \rightarrow Z_1 = v_{11}X_1 + v_{12}X_2 + \cdots + v_{1p}X_p$$

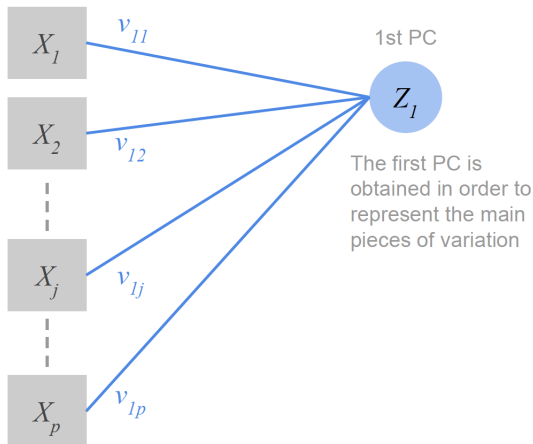
$$PC_2 \rightarrow Z_2 = v_{21}X_1 + v_{22}X_2 + \cdots + v_{2p}X_p$$

$$\vdots$$

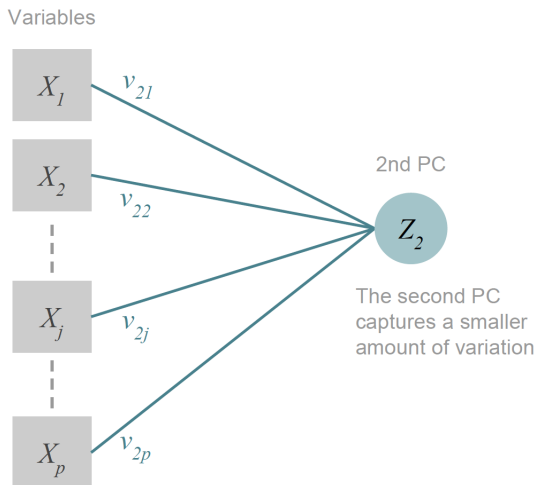
$$PC_r \rightarrow Z_r = v_{r1}X_1 + v_{r2}X_2 + \cdots + v_{rp}X_p$$

PCA AS BEST SUMMARY OF INFORMATION

Variables

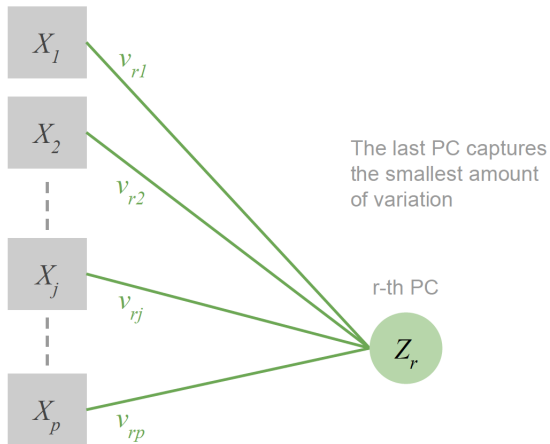


PCA AS BEST SUMMARY OF INFORMATION

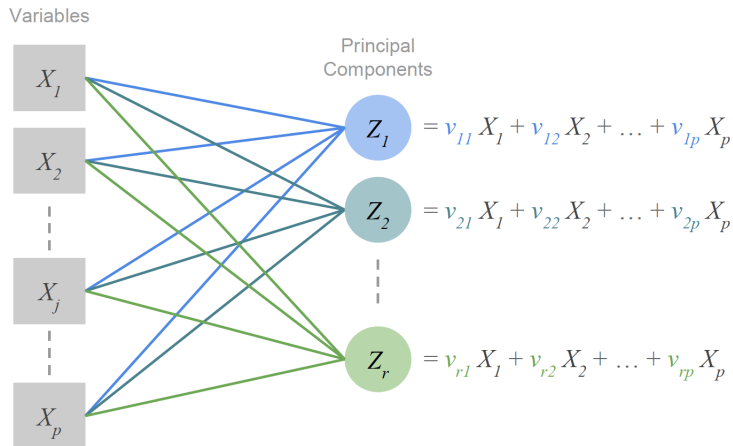


PCA AS BEST SUMMARY OF INFORMATION

Variables



PCA AS BEST SUMMARY OF INFORMATION



PCA AS BEST SUMMARY OF INFORMATION

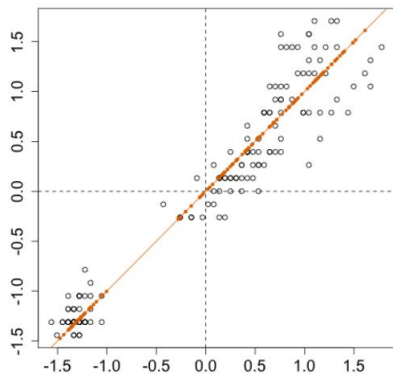
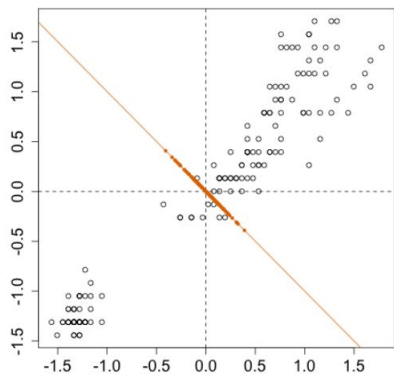
Main requirement

The main requirement for the PCs, $Z_1, \dots, Z_r$, is that they need to capture the most variation in the data X

Convenient requirement

To avoid a PC capturing the same variation as other PCs (i.e. avoiding redundant information), we may also require them to be mutually orthogonal, so they are uncorrelated with each other.

PCA AS BEST SUMMARY OF INFORMATION



PCA AS BEST SUMMARY OF INFORMATION

Finding the directions with maximum variance means, algebraically:

- Maximizing a quadratic form which involves the var-cov matrix of the centered data S .
- The solution is given by the spectral decomposition (diagonalization) of S
- The eigenvalues of S are related to the amount of variance explained by each component. Assume $(\lambda_1, \dots, \lambda_r)$ are the eigenvalues of S :

$$\left(\begin{array}{c} \text{Proportion of total} \\ \text{Variance due to} \\ \text{ith principal} \\ \text{component} \end{array} \right) = \frac{\lambda_i}{\lambda_1 + \dots + \lambda_r}$$

PCA AS BEST SUMMARY OF INFORMATION

PCs in matrix format:

$$Z_{n,r} = X_{n,p} V_{p,r}$$

- PCs are just linear combinations of X s. The weights V are given by the normalized eigenvectors of $S = X'X$, say $(\lambda_1, e_1), \dots, (\lambda_r, e_r)$, for eigenvalue-vector pairs.
- We can obtain as many PCs as the rank of X (i.e. $\text{rank}(X) = r$).
- The principal components are uncorrelated.
- They have variances equal to the eigenvalues of S .

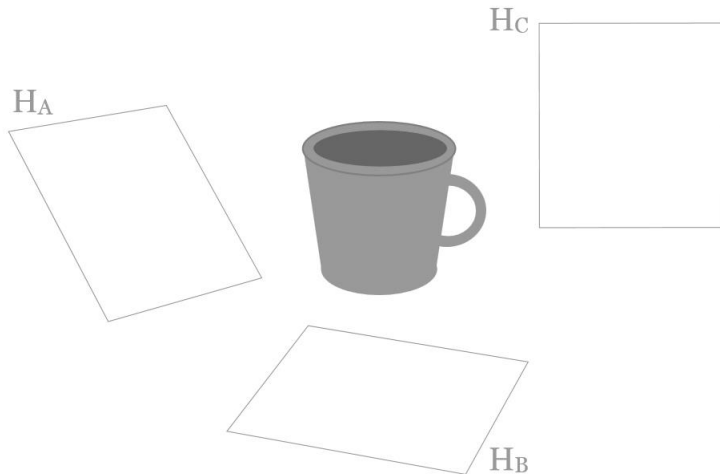
GEOMETRIC APPROACH

Some data in some high dimensional space:



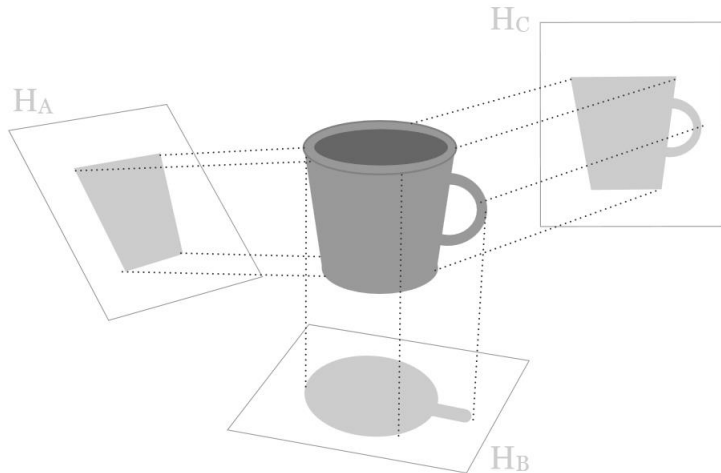
GEOMETRIC APPROACH

We are looking for candidate subspaces



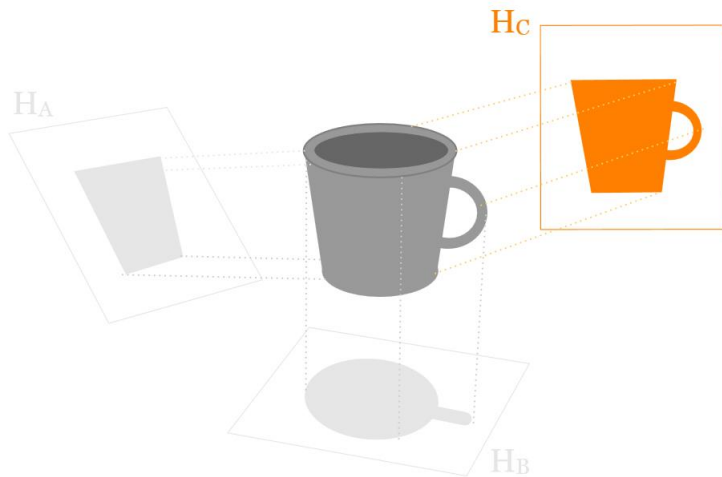
GEOMETRIC APPROACH

... with the best low-dimensional representation ...



GEOMETRIC APPROACH

maximizing projected inertia

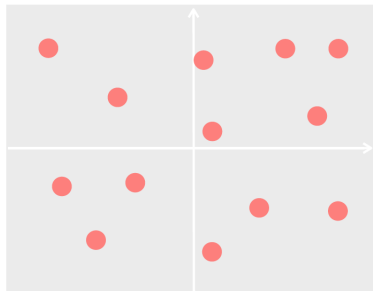
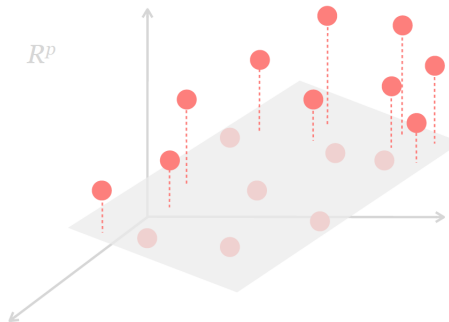


Looking at the cloud of points:

- Under a purely geometric approach, PCA aims to represent the cloud of points in a space with reduced dimensionality in an optimal way, that is, making the dispersion of the projected points as large as possible.
- We look for the best graphical representation that allows us to visualize the cloud of individuals in a low dimensional space (usually 2-dimensions).

GEOMETRIC APPROACH

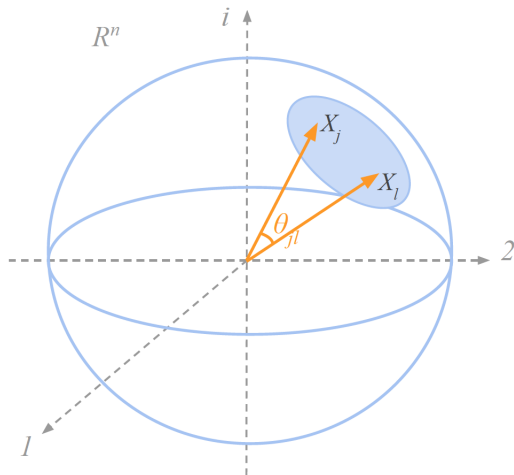
Looking at the cloud of points:



How do we find the associated axes?

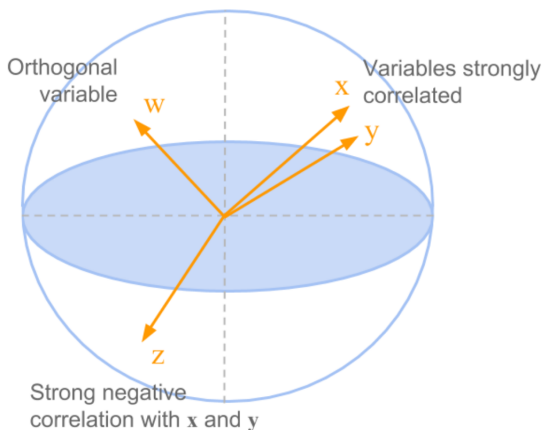
GEOMETRIC APPROACH

Looking at the cloud of standardized variables. All the variables are located within a one-unit distance from the origin (on a hypersphere of radius 1).



GEOMETRIC APPROACH

Looking at the cloud of standardized variables



GEOMETRIC APPROACH

Looking at the cloud of standardized variables:

- With standardized data, the p variables are located within a hypersphere of radius 1 in an n -dimensional space.
- We represent them graphically as vectors.
- The scalar product between two variable X_j and X_l is:

$$\langle X_j, X_l \rangle = \sum_{i=1}^n x_{ij} x_{il} = \|x_j\| \|x_l\| \cos \theta_{jl}$$

- Notice that:

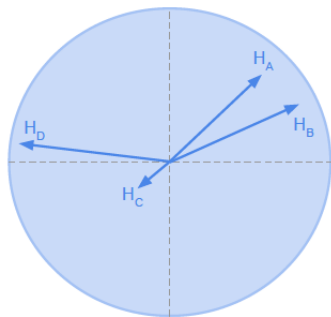
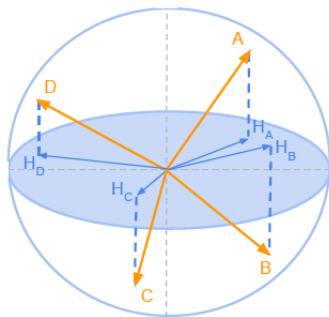
$$\cos \theta_{jl} = \frac{x_j^T x_l}{\|x_j\| \|x_l\|} = \text{cor}(X_j, X_l)$$

Projecting the cloud of standardized variables:

- The property $\cos \theta_{jl} = \text{cor}(X_j, X_l)$ is essential in PCA.
- A representation of the cloud of variables can be used to visualize correlations (through the angles between variables).
- The cloud of variables is also projected onto a low dimensional space.
- In this case, the distance between two variables is computed with inner products.

GEOMETRIC APPROACH

Projecting the cloud of standardized variables:



Projecting the cloud of variables on the main plane of variability. We seek to find a projection plane providing the largest amount of information about the associations between the variables, i.e., angles between the variables.

GEOMETRIC APPROACH

Projecting the cloud of standardized variables:

- The projection of variable j onto an axis k , is equal to the cosine of the angle θ_{jk} .
- The criterion maximizes:

$$\sum_{j=1}^p \cos^2 \theta_{jk} = \sum_{j=1}^p \text{cor}^2(x_j, z_k)$$

where z_k is the new variable which is the most correlated with all the original variables.

PCA AS OPTIMAL APPROXIMATION OF DATA

The role of *Models* in Statistical Analysis,

$$Data = Structure + Residual$$

In PCA:

- We can obtain as many PCs as the rank of X (i.e. $rank(X) = r$).

$$\underset{(n,p)}{X} = \underset{(n,r)}{Z} \underset{(r,p)}{V'}$$

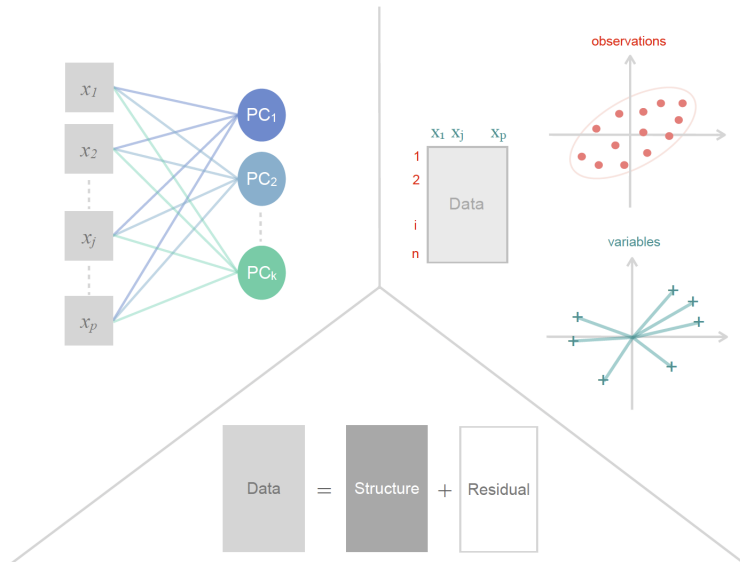
- But usually, we will retain only just a few PCs ($m \ll p$).

$$X_{n,p} \approx Z_{n,m} V'_{m,p} + Residual$$

$$X_{n,p} \approx Z_{n,m} V'_{m,p} = \hat{X}$$

where Z is the matrix of PCs or scores and V is the matrix of eigenvectors. Just a few PC's will optimally summarize the main structure of the data.

SUMMARY: THREE PERSPECTIVES FOR PCA



MINIMAL OUTPUT...

.. from any PC analysis:

Eigenvalues provide information about the amount of variability captured by each principal component

Scores or PCs provide coordinates to graphically represent objects in a lower dimensional space.

Loadings provide the correlation between variables and components. They often help to interpret the components as they measure the contribution of each individual variable to each principal component. However, they do not indicate the importance of a variable to a component *in the presence of others*.

Eigenvectors are unit-scaled loadings. They are the coefficients of orthogonal transformation (rotation) of variables into PC's.

SOME QUESTIONS...

- How many principal components should we retain?
- How good (or bad) is the representation of the original data in terms of the retained components?
- What variables characterize each component?
- Which variables are influential and how are they correlated?
- Are there any outlier observation?

COMPONENTS TO RETAIN

There is not universal agreement. Some informal, ad hoc rules of thumb:

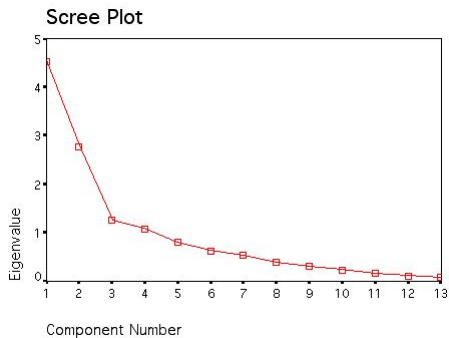
- Include just enough components to explain a predetermined portion of variation, for instance 70 % or 90 %. These percentages will generally become smaller as p or n increases.
- Exclude those components whose eigenvalues are less than the average, i.e., less than one if the components have been extracted from the correlation matrix.

COMPONENTS TO RETAIN

- Look for an *elbow* in the curve plotted in the *scree plot*. This point is considered to be where large eigenvalues cease and small eigenvalues begin.
- Kaiser's rule, which consists of retaining those PCs with eigenvalues $\lambda_k > 1$.
- Jolliffe's rule, in which we retain those PCs with eigenvalues $\lambda_k > 0.7$

Modern research attempts to find better thresholds for square and rectangular data matrices. In [this article](#) you can find a table of approximate values for a threshold: empirical singular values below that threshold are set to 0.

SCREE PLOT



Note that the usefulness of the principal components is two-fold:

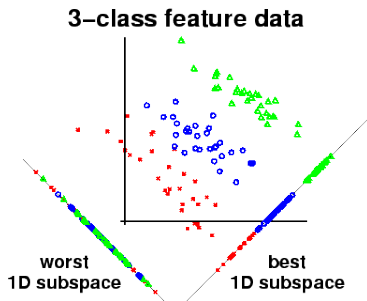
- It allows for an optimal representation, in a space of reduced dimensions, of the original observations. In this way, PCA is the first step towards identifying possible latent variables that may have generated the data.
- It allows the original correlated variables to be transformed into new uncorrelated variables, facilitating the interpretation of the data.

Warnings:

- PCA is a descriptive technique, no assumptions are made about a model or distribution where the data come from. All it says is *these data can be summarized using projections along these dimensions*, retaining a percentage of the total system variability.
- Data is reduced in terms of how many new variables are retained, not how much data has to be collected (all the original variables are required).
- It's general a good idea to standardized all the variables first, although sometimes it is not strictly necessary. We will do that when variables are measured in different scales and they have very different variances.

MORE INFORMATION ...

- PCA plots may reveal a tendency of points to cluster, i.e., groups of points. But they are not designed to account for groups within the data. If separate groups exist, they may be hidden. The space spanned by the vector associated with the first principal components, will not necessarily be the best for discriminating groups.



MORE INFORMATION ...

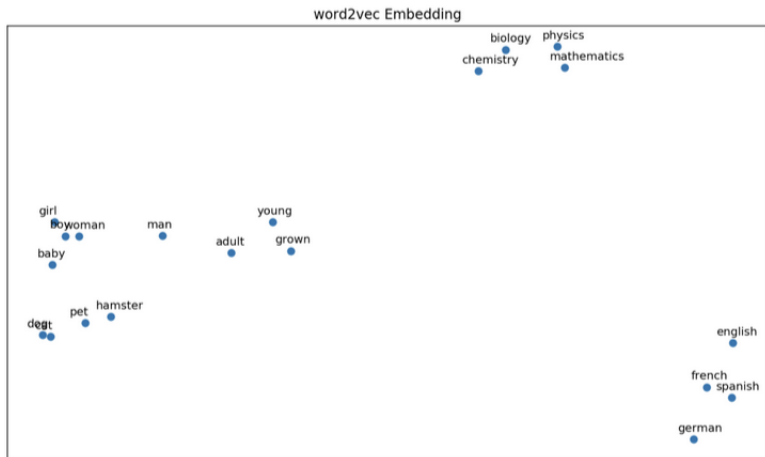
- To get interpretable results, variables should be linearly related, or at least, monotonically related to one another. It won't work if variables are already uncorrelated with each other.
- They can reveal, as well, multivariate outliers. In fact, many methods for outliers detection are based on PCA.
- They can be used as independent vars. to solve multicollinearity issues in Multiple Linear Regression.

PCA is a flexible approach and it is a means to a end rather than an end in itself, because it frequently is an intermediate step in much larger analyses.

Related Methods:

- Principal coordinate analysis or Metric Multidimensional scaling
- Correspondence Analysis
- Non Metric Multidimensional scaling
- Canonical Variate Analysis
- Factor Analysis
- Independent Component Analysis (ICA), a potentially powerful technique that seeks to uncover hidden variables in high-dimensional data. It is technically demanding and also rather specialised, but it is available in R via the add-on package **fastICA** (Marchini, Heaton and Ripley, 2010). Some areas where it is used:
 - Multichannel Electrocardiogram Analysis, separating signals.
 - Monitoring human brain-wave activity.
 - Functional Magnetic Resonance Imaging experiments.

VISUALIZING WORD EMBEDDINGS



JOKE OF THE WEEK

